



Effect of Laser Positioning on the Microstructure and Properties of NiTi-Copper Dissimilar Laser Welds

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Dissimilar pulsed laser welding of 400- μm NiTi and Cu wires was used to investigate the effect of laser positioning on NiTi-Cu joint performance. Laser positioning was found to have a significant role in the microstructure due to the difference in mixing patterns and composition distributions. Homogenous element distribution was observed in both the NiTi offset and the centerline joints, while complex mixing patterns were observed in the Cu offset joints. Changing the laser position resulted in a change in alloy content in the weld pool that affected the mixing patterns, and the phases that were formed in the weld zone of each sample. By moving the laser beam from NiTi toward Cu, NiTi and NiTiCu in the weld zone were replaced by pure Cu which resulted in a decreased average hardness from ~ 512 to ~ 158 HV. However, high hardness values were found inside the weld zone of Cu offset welds due to the presence of hard and brittle ternary intermetallic compounds. The 100- μm Cu offset joint displayed the worst mechanical response due to its inhomogeneous microstructure and the presence of cracks and pores. It was shown that positioning the laser beam on the NiTi or on the centerline resulted in a better transition in microstructure that led to acceptable mechanical properties without optimizing laser parameters or inserting any interlayers.

Keywords copper, dissimilar welding, laser positioning, laser welding, NiTi alloy, shape-memory alloys

1. Introduction

NiTi shape-memory alloys (SMAs) are one of the most widely used smart materials with applications in aerospace, micro-electronics and medical devices due to their superior pseudoelasticity (PE) and shape-memory effect (SME) (Ref 1–3). In many of these applications, the appropriate similar and even dissimilar NiTi joining is needed to manufacture complex multi-component systems (Ref 4, 5). In comparison with other joining methods, laser welding has attracted more attention for joining NiTi SMAs for a variety of shapes such as wires, sheets and tubes (Ref 6–9) due to its high energy concentration, high precision, rapid processing potential and flexibility for automa-

tion (Ref 10–12). Thus, most of the published studies for dissimilar joining of NiTi focused on the laser welding (Ref 13–18). However, these dissimilar joints are extremely sensitive to the formation of hard and brittle intermetallic compounds (IMCs) in the weld zone, which dramatically deteriorate the joint performance (Ref 5).

There are many published methods for controlling the formation of intermetallic compounds, which include using interlayers between two base materials (Ref 19, 20) or optimizing laser parameters to minimize the mixing of elements (Ref 21). Some studies showed the influence of laser spot position on the mixing of elements, microstructure and properties of the weld zone. For instance, Panton et al. (Ref 17) studied the effect of laser positioning on the dissimilar welding of NiTi and MP35N. It was revealed that the composition of the weld zone was affected by laser positioning and macroscopic cracks were observed when the laser was positioned on the NiTi side. Oliveira et al. (Ref 22) carried out the dissimilar laser welding of NiTi and Ti6Al4V with niobium interlayer. When positioning the laser on the Ti6Al4V side of the joint, joining occurred with a diffusive interlayer on the Ti6Al4V–Nb interface, while eutectic formation occurred on NiTi–Nb side of the joint. No joint formed when the laser was positioned on the NiTi side of the joint.

There has been an increasing demand for the production of NiTi-based medical devices (pacemakers and implants) which have a required electrical conductivity while preserving NiTi functional properties (Ref 14, 18). Therefore, dissimilar welding of NiTi to a cost-effective and highly conductive material such as Cu would be an ideal solution (Ref 21). Cu is widely used in electronic interconnections due to its high electrical and thermal conductivity, high ductility and corrosion resistance (Ref 23, 24). Although laser welding is an ideal process for joining Cu (Ref 25), copper's low absorptivity to lasers leads to the high dependency of joint performance on the

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thermal transfer from the less reflective metal during dissimilar welding of Cu and other alloys (Ref 23).

Among notable studies done on the laser welding of NiTi-Cu, Zeng et al. (Ref 21, 24) studied the effect of peak power on joint performance in the dissimilar welding between NiTi wire and Cu sheets. Increasing the power of the laser increased mixing of the NiTi and Cu alloys, which resulted in higher hardness and greater amounts of cracking localized along the interface of Ti-rich and Ti-lean regions. Achieving acceptable joints required a time-consuming laser parameter optimization process (Ref 24). In addition, most of the industrial applications limit the joint configuration (Ref 5). Laser positioning welding is a promising solution to overcome these challenges as mentioned earlier in the joining of NiTi and Mp35N (Ref 17) and other studies of dissimilar laser welding (Ref 26-28). This is an effective method since laser offset welding could provide different patterns of element mixing, and therefore, different phases could be formed and properties would be highly influenced by moving the laser beam.

Nonetheless, to the authors' best knowledge, there are no published reports on the effect of laser offset welding on dissimilar NiTi and Cu joints. Laser offset welding can be used to control the mixing of Ni, Ti and Cu to control the weld metallurgy with the aim of improving the mechanical and microstructural properties, as shown in NiTi-stainless steel joints in the authors' recent publication (Ref 29). Thus, this present work may open the door for fabricating biomedical devices by studying effects of laser beam offset on the microstructure, mixing patterns and mechanical properties of NiTi-Cu dissimilar butt-welded joints.

2. Materials and Methods

A superelastic Ni-rich NiTi (50.2 at.% Ni) wire with 400 μm diameter and commercially pure Cu (99.9 at.%) wire with the same diameter were used in this present study. The wires were cleaned with acetone, ethanol and de-ionized water to remove any impurities and contamination before laser welding.

Laser welding was carried out using the Miyachi Unitek LW50A pulsed Nd:YAG laser, with a wavelength of 1.064 μm and nominal spot size of 400 μm . Pure argon was selected as a shielding gas, with a flow rate of 30 cubic feet per hour (cfh). The top-hat-shaped laser profile with a 0.9 kW peak power and width of 15 ms, including 1 ms upslope and 11 ms downslope, was used for all experiments. The laser was positioned on five different locations on the NiTi side, centerline and Cu side which are depicted in Fig. 1. Focusing the laser on NiTi with a 100- μm offset could not make a sound joint with Cu and therefore will not be discussed further. X-ray diffraction (XRD)

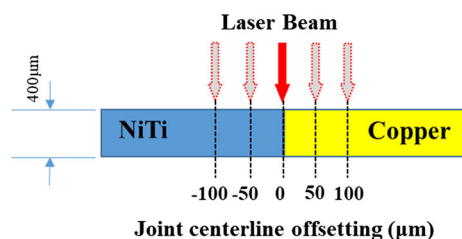


Fig. 1 Schematic of the laser welding process with different laser offsets

measurements were taken at room temperature on a Bruker D8 diffractometer with Co source with the scanning range of $2\theta = 30^\circ$ - 100° .

Cross sections of welds were mounted in epoxy and then grinded with sand paper up to number 1200 and then polished to 1 μm using diamond sprays, followed by etching with Kroll's reagent for 30 s. The related microstructures were observed using Olympus BX51M Optical microscope (OM) and Zeiss Ultra Plus field emission scanning electron microscope (SEM) equipped with energy-dispersive spectroscopy (EDS) to analyze the compositional changes. A Clemex CMT automated micro-Vickers hardness tester was used to make a series of 50 g indents across the base materials and weld zone, 50 μm apart with a holding time of 10 s. Tensile testing was performed four times for each weld with a gauge length of 12 mm and a strain rate of $3 \times 10^{-4} \text{ s}^{-1}$ using an Instron 5548 micro tester.

3. Result and Discussion

3.1 Weld Zone Appearance

Figure 2, 3 and 4 displays the cross sections of NiTi-Cu dissimilar joints welded by different laser offset positions and their corresponding elemental distribution map. The weld width decreased when moving the laser position from NiTi to Cu, and the mixing pattern inside the weld zone changed significantly. The weld zone in the NiTi offset and centerline welds (Fig. 2 and 3) have uniform element mixing. The Cu offset welds (Fig. 4 and 5) have a complex mixing pattern and elemental distribution, which will be discussed in the next section. Focusing the laser on the NiTi side (Fig. 2a) resulted in forming the largest weld width of 1150 μm . As the laser was positioned toward the Cu side of the joint, the weld width decreased to 950 μm (Fig. 3a), 500 μm (Fig. 4a) and 375 μm (Fig. 4e).

It is reported in dissimilar welding between Cu and other alloys that Cu was melted not by the welding heat source, but by heat transfer from the melted dissimilar material (Ref 27). This is due to copper's high thermal conductivity, higher melting point and expansion coefficients (Ref 27, 28). In the current study, NiTi has lower thermal conductivity and laser reflectivity than Cu (Table 1); therefore, most of the input energy for melting the joint was through absorption of the laser by the NiTi and heat transfer to Cu. The Cu was melted by the liquid NiTi alloy (Ref 20, 28). As the laser offset changed from the NiTi to the Cu side, the energy absorption decreased, leading to a decreased weld pool width.

3.2 Microstructure and Elemental Distribution

Figure 5 presents the magnified microstructure inside the weld zone in different samples labeled by A, B, C and D in Fig. 2 and 3 and 4(a) and (e). The welds with the laser offset onto the NiTi ($-50 \mu\text{m}$ in Fig. 2a and 5a) and on the centerline ($0 \mu\text{m}$ in Fig. 2a and 5b) had columnar dendritic microstructures. This dendritic solidification microstructure is formed due to higher mixing of elements inside the weld zone (Ref 20, 24). The weld with the laser offset onto the Cu had a variety of microstructures with complex elemental mixing patterns (Fig. 5c and d). With increasing Cu content in the weld zone, cellular and dendritic microstructures were suppressed (Ref 20). EDS point analysis was performed on the labeled

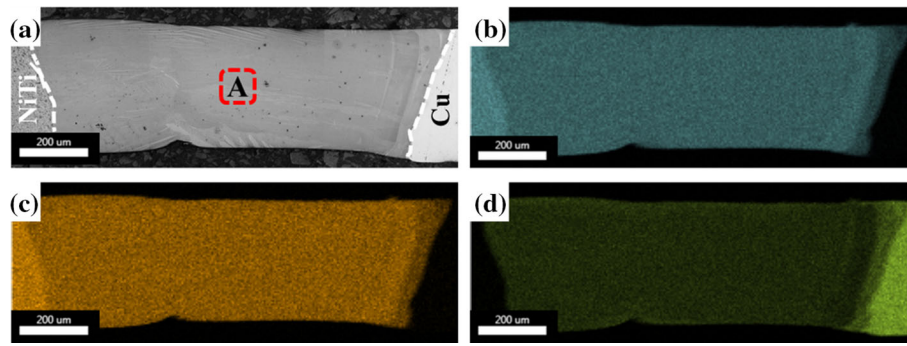


Fig. 2 (a) Weld zone microstructure of the laser focused on the NiTi ($-50\ \mu\text{m}$) and its EDS elemental map for (b) Ti, (c) Ni and (d) Cu. A in (a) is selected in the middle of weld zone and will be studied later

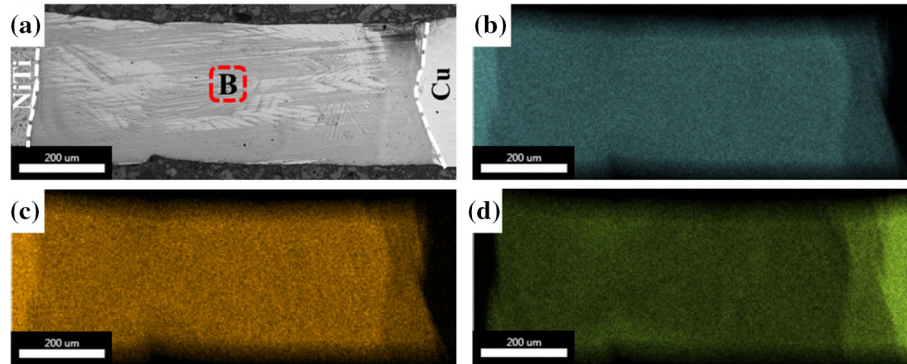


Fig. 3 (a) Weld zone microstructure of the laser focused on the centerline and its EDS elemental map for (b) Ti, (c) Ni and (d) Cu. B in (a) is selected in the middle of weld zone and will be studied later

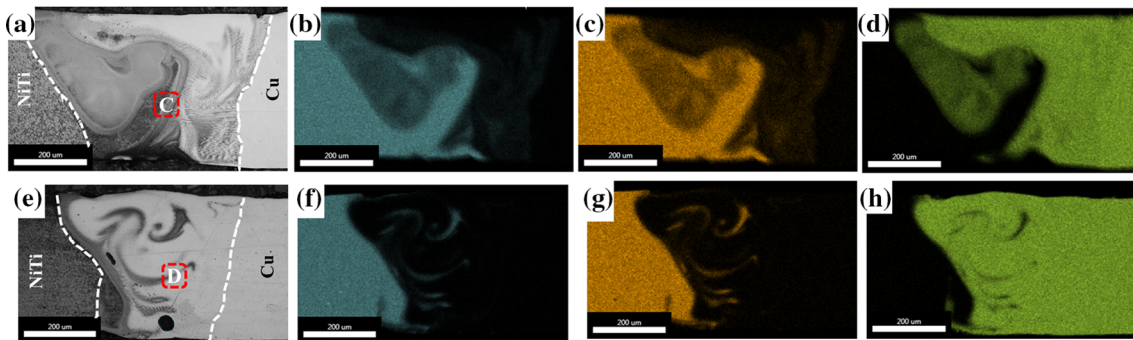


Fig. 4 Weld zone microstructure of the laser focused on (a) Cu ($50\ \mu\text{m}$) and (e) Cu ($100\ \mu\text{m}$) and their EDS elemental map for (b) and (f) Ti, (c) and (g) Ni and (d) and (h) Cu. C and D in (a) and (e) are selected in the middle of weld zone and will be studied later

points on different zones (P1 to P6 in Fig. 5c and P1 to P7 in Fig. 5d), and the results are listed in Table 2. The microstructures in region C (Fig. 5c) consisted of zones I, II and III which are related to gradual changes from dendrites to lamellar microstructures from left to right. The EDS results (Table 2) and Ni-Ti-Cu alloy ternary phase diagrams (Ref 30) indicated that NiTi and Ni-Ti IMCs may be present in zone I. Thus, it can be inferred that the re-melting and solidification of NiTi base metal occurred in this region. As previously mentioned, Cu has much higher heat conductivity and reflectivity, in addition to a higher melting point. The higher conductivity, temperature and reflectivity of Cu compared to NiTi increased the cooling rate of the weld, resulting in the entrapment of NiTi in the weld

zone with insufficient time for mixing with Cu. In zone II (Fig. 5c), the Cu content increased from 22.74 at.% to 68.06 at.% from P3 to P5. The existing phase in this region could be a mixture of $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ hard and brittle phase, Ni-Ti IMCs and Cu-Ti IMCs (Ref 31-33). In zone III, the Cu content is more than ~ 80 at.%, and there is a potential for dissolution of Ni and Ti in the Cu and formation of a Cu solid solution (Ref 34, 35). In the case of the 100- μm offset on the Cu side (Fig. 5d), different zones with chemical compositions related to the selected points were characterized (Table 2). Pure Cu was detected in the weld zone (VI zone) which was not present in the former condition (50- μm offset on Cu).

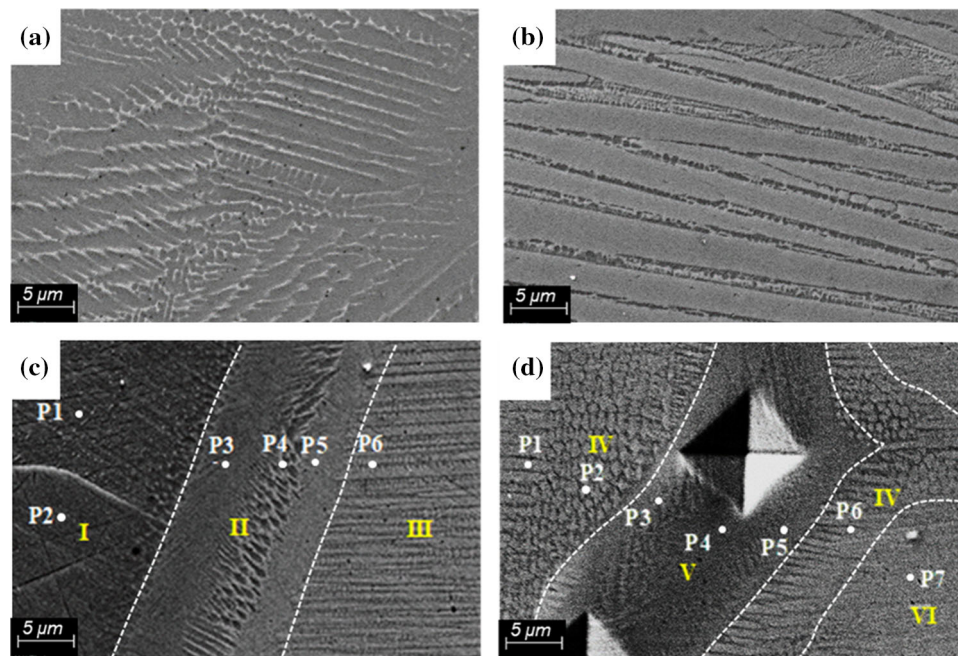


Fig. 5 Weld zone microstructure from different welded conditions in NiTi-Cu joints where laser spots pointed on (a) NiTi ($- 50 \mu\text{m}$), region A in Fig. 2(a) and (b) centerline ($0 \mu\text{m}$), region B in Fig. 3(a) and (c) Cu ($50 \mu\text{m}$), region C in Fig. 4(a) and (d) Cu ($100 \mu\text{m}$), region D in (a)

Table 1 Thermo-physical properties of experimental materials at room temperature

Metal	Melting point, °C	Conductivity, W/m °C	Heat capacity, J/(kg °C)	Absorptivity at $1 \mu\text{m}$ laser, %
NiTi (Ref 17, 29)	1310	10	77	~ 35
Copper (Ref 18)	1083	401	386	3

Table 2 Composition and possible phases in selected points in the weld zone shown in Fig. 5

Samples	Zone	Points	Cu, at. %	N, at. %	Ti, at. %	Possible phases (Ref 31-35)
NiTi ($- 50 \mu\text{m}$)	A	...	21.7	39.1	39.2	NiTiCu matrix and probable $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ IMCs
Centered ($0 \mu\text{m}$)	B	...	31.4	33.0	35.6	
Cu ($50 \mu\text{m}$)	I	P1	0	55.9	44.1	NiTi islands
		P2	0	55.9	44.1	
	II	P3	22.7	41.9	35.4	NiTiCu matrix and probable $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ IMCs
		P4	52.5	23.8	23.8	
		P5	68.1	15.6	16.4	
	III	P6	80.3	9.5	10.2	Copper solid solution
Cu ($100 \mu\text{m}$)	IV	P1	83.6	7.9	8.4	NiTiCu matrix and probable Cu-Ti and $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ IMCs
		P2	75.3	12.2	12.5	
	V	P3	55.5	23.0	21.6	
		P4	35.4	34.7	29.9	
		P5	42.2	31.2	26.6	
	IV	P6	82.5	8.4	9.2	
	VI	P7	100	0	0	Copper solid solution
						Pure copper

The reason for the presence of pure Cu inside the weld zone is the fact that the Cu content was greater than the other offset conditions, as well as the high thermal conductivity of Cu and laser rapid cooling rate, leading to an uneven mixing of Cu with

the NiTi alloy. Measuring the chemical composition of selected points in zones VI and V and comparing them with the Ni-Ti-Cu ternary phase diagram lead to a belief for the presence of Cu solid solution and in some extent Ni-Ti IMCs and Cu-Ti IMCs

as possible phases (Ref 20, 32). To identify the formed phases in the welds, x-ray diffraction (XRD) was adopted. The XRD graph is plotted for all the samples with different laser positions and can be seen in Fig. 6. Overall, various phases are formed inside the weld zone of each sample mainly consisting of NiTi, pure Cu and equiatomic NiTiCu phase in two different forms (NiTiCu and Ni_{0.37}Ti_{0.33}Cu_{0.3}). By positioning the laser on Cu, a phase consisting mainly of Cu is formed inside the weld zone and some NiTiCu phases are present. As seen, in the centerline condition and positioning on Ni (− 50 μm), most of

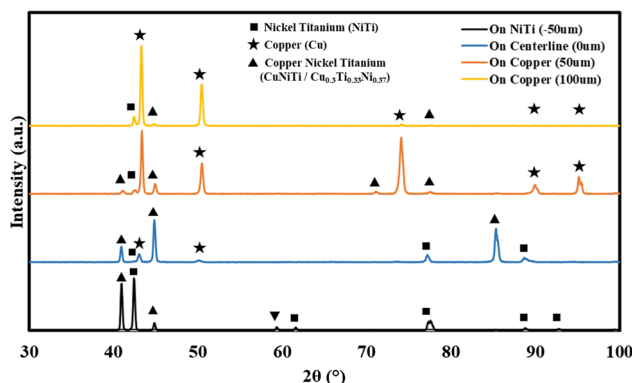


Fig. 6 X-ray diffraction of welded samples showing different phases of the different laser positions on the NiTi/Cu joint

the peaks are related to NiTi and NiTiCu phases which is in good agreement with Table 2.

3.3 Hardness

Figure 7 presents the microhardness profile mapping of dissimilar welded samples with different laser offsets. As observed, the Cu base material has a significantly lower hardness (90 HV) and NiTi base material is the harder base material in this joint (~ 380 HV). However, in almost all joints, the weld zone has higher values of hardness due to mixing of different elements of the base materials and formation of new phases and hard intermetallic compounds (mainly Ni_xTi_yCu_z). As mentioned before, with changing the offset from NiTi to Cu, the weld zone width decreases drastically and therefore, the higher hardness region decreases; the NiTi offset condition (Fig. 7a) has large and extensive high hardness regions, and the Cu offset samples (Fig. 7c and d) have very small and narrow high hardness regions in the weld zone. On the other hand, as expected from EDS maps (Fig. 2, 3 and 4), positioning the laser on NiTi or on the centerline resulted in more homogenous weld zones both in chemical composition and hardness values. Nevertheless, the hardness values inside the weld zone in Cu offset samples are uneven and experienced near 80 to 450 HV in some points. Based on EDS maps and point scans, this variation in hardness pointed to complex mixing patterns of Cu inside Ti and Ni elements which caused the formation of pure Cu islands (indicated by the color green in Fig. 4d, h, 7c and d) in some parts and Ni_xTi_yCu_z

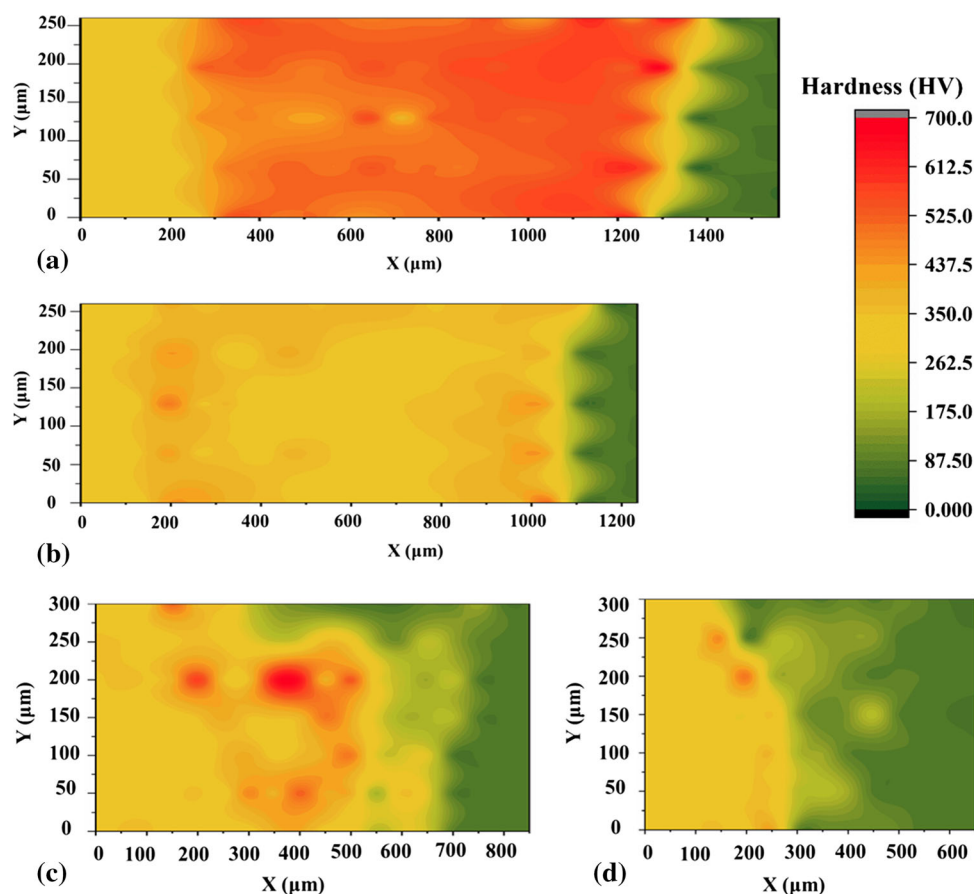


Fig. 7 Microhardness mapping of NiTi-Cu welded joint offsetting (a) 50 μm on NiTi, (b) on base metals centerline (c) 50 μm on Cu and (d) 100 μm on Cu

intermetallic hard phases (Table 2 and red color in Fig. 7c and d). It is worth noting that the hardness of pure Cu was measured 105 ± 4 HV; therefore, the green area (hardness less than 100 HV) is a sign of softness in the Cu side HAZ (Ref 36).

To compare the hardness values quantitatively, the average hardness values inside the weld zones are depicted in Fig. 8. It could be observed that with changing the offset from NiTi ($-50 \mu\text{m}$) to Cu ($+100 \mu\text{m}$), the average hardness values decreased from ~ 512 to ~ 158 HV, respectively. The main reason for higher hardness in the NiTi offset condition is a greater mixing of elements and an increased possibility of the presence of $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ which is a hard and brittle phase [especially when z is higher than 10% (Ref 31)].

By altering the focus of the laser onto the centerline and then Cu, the weld zone had less Ti and more Cu content, which

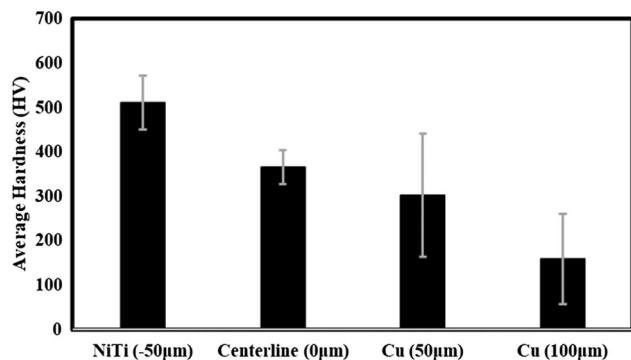


Fig. 8 Effect of laser position offset on average hardness inside the weld zone of NiTi-Cu

means the probable presence of Cu-Ti IMCs (the softer IMCs) instead of NiTiCu and Ni-Ti IMCs. Thus, it can be considered that the hardness of the weld zone decreased with shifting the laser position from the NiTi side to the Cu side, although some local regions in these samples could be found with high hardness values due to Ti-rich and Cu-lean regions.

3.4 Tensile Properties

Figure 9 shows the stress-strain curves and the locations of failure for all of the discussed joints. The extracted mechanical properties from these stress-strain curves are shown in Table 3. The joints with offset onto the NiTi ($-50 \mu\text{m}$) and the centerline ($0 \mu\text{m}$) had the highest tensile strengths (Fig. 9a), and fracture occurred in the Cu heat-affected zone (Fig. 9b and c). In comparison with these conditions, the tensile strength of the Cu ($50 \mu\text{m}$) offset joint decreased slightly, but with a near similar extension and fracture location (Fig. 9d). On the

Table 3 Mechanical properties of as-received and welded samples extracted by tensile stress-strain curves shown in Fig. 9

Sample	YS, MPa	UTS, MPa	Elongation, %
NiTi	915.1 ± 5.1	1257 ± 4.8	21.8 ± 0.3
Cu	179.1 ± 6.4	245 ± 2.4	47.5 ± 0.4
NiTi ($-50 \mu\text{m}$)	136.8 ± 6	216.9 ± 3.2	11.8 ± 1.5
Centerline ($0 \mu\text{m}$)	155.8 ± 7.1	234.7 ± 4.1	12.1 ± 0.8
Cu ($50 \mu\text{m}$)	148.8 ± 6.5	229.9 ± 2.4	12.2 ± 1.2
Cu ($100 \mu\text{m}$)	175.6 ± 7.2	175.6 ± 7.8	6.4 ± 1.1

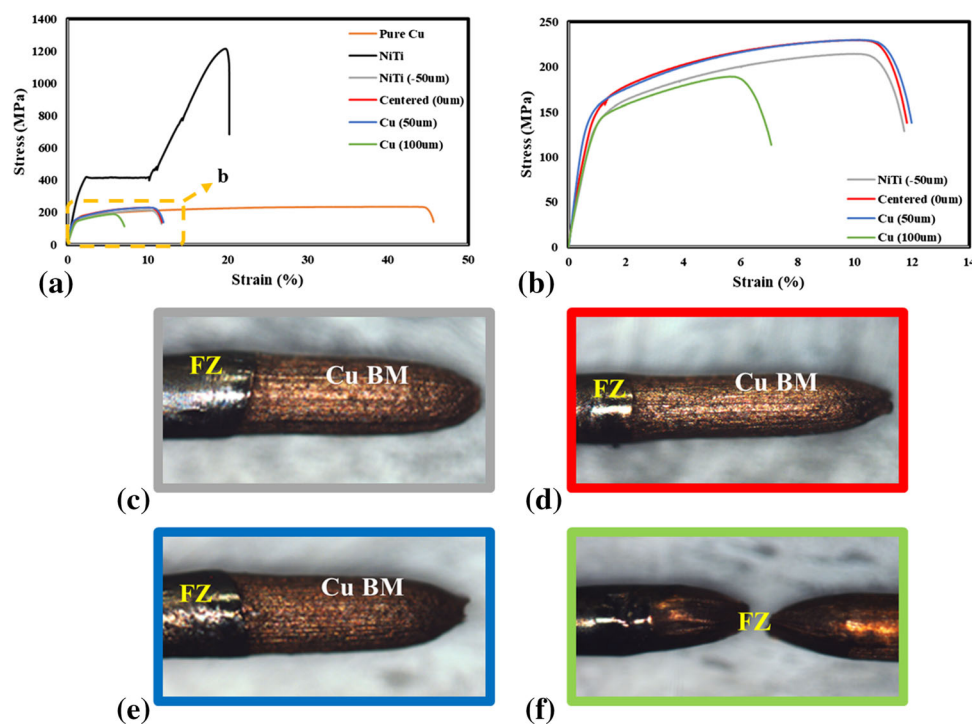


Fig. 9 (a) Stress-strain curves for all samples, (b) stress-strain curves for welded samples and the corresponding fracture locations for (c) offset on NiTi ($-50 \mu\text{m}$), (d) centerline ($0 \mu\text{m}$), (e) $50\text{-}\mu\text{m}$ offset on Cu and (f) $100\text{-}\mu\text{m}$ offset on Cu

contrary, more offset on Cu (100 μm) resulted in a drastic decrease in both the tensile strength (more than 50 MPa) and the extension (6%) (Fig. 9a and Table 3). It is also visible that the fracture location in the 100- μm offset sample changed from the Cu base material to the weld zone of the NiTi-Cu joint (Fig. 9e). It should be mentioned that compared to the Cu (100 μm) joint, the mixing pattern of the Cu (50 μm) joint was also not uniform, but the defect-free weld zone ensured the properties of the joints and the fracture occurred in the softer Cu base metal. Moreover, as the Cu (100 μm) sample absorbed more energy into the Cu side of the joint, the heat-affected zone of the Cu would have bigger grain sizes (Ref 36) and the effect of HAZ on the deterioration of the Cu(100) strength should not be neglected.

To have a better understanding of the differences between these fracture locations, fracture surfaces of offset – 50 μm on NiTi (with failure from the Cu base material) and the 100- μm Cu joint (failure occurring at the weld zone) are depicted in Fig. 10. The other two samples (centerline and 50- μm offset on Cu) have the same mechanical properties (Fig. 9), the same fracture locations (Fig. 9 shows Cu base material as fracture location) and almost the same fracture surface features. The top view and low-magnification fracture surfaces of the aforementioned welded samples are depicted in Fig. 10a and d. The detailed view of the magnified fiber zone of the welded samples is seen in Fig. 10. Fracture surfaces of the first three mentioned samples (the candidate is NiTi – 50 μm in Fig. 9a-c) display many deep and equiaxed dimples in the fiber region, implying ductile fracture in the Cu base material (Ref 37, 38) (shown in Fig. 10b and c). However, with the laser position focused on Cu at 100 μm (Fig. 10c and d), the fiber region shrinks

(Fig. 10d) and the details inside the fiber region indicate tear ridges coexisting with decreasing amount of dimples and pores which indicate a more brittle nature of this welded sample (Ref 37, 38).

The microstructure of the Cu (100 μm) contains cracks and pores inside the weld zone as shown in Fig. 11. Some points surrounding these defects were selected for analyzing the composition by using EDS, and the results are gathered in Table 4. There are sharp gradients in the composition (Table 4) for different sides of the crack (P1 and P2 in Fig. 11), which could indicate that the crack is located at the interface of the Cu solid solution and brittle $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ IMCs. Previous studies showed that the formation of cracks in dissimilar NiTi welds had been attributed to both solidification cracking and brittle intermetallic compounds (Ref 17, 25). It should be mentioned that the crack appears at the interface of Cu-lean (P1 in Fig. 11a) and Cu-rich (P2 in Fig. 11b) regions. Therefore, due to the formation of these different phases, solidification cracking has great potential to form (Ref 20, 24). Additionally, the chemical composition of P1 in Fig. 11(a) has a great potential to form $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ hard and brittle intermetallic compounds and formation of cracks would be inevitable.

Formation of pores in the Cu offset weld sample (Fig. 10b) referred to the high conductivity and fluidity of Cu combined with the short cooling time of the laser process. Therefore, any created bubble inside the welding pool had no time to escape and pores could be formed during solidification (Ref 25). Consequently, it can be concluded that focusing the laser on Cu increases the potential for the formation of cracks and pores inside the weld zone and will deteriorate the mechanical properties of NiTi-Cu dissimilar joints.

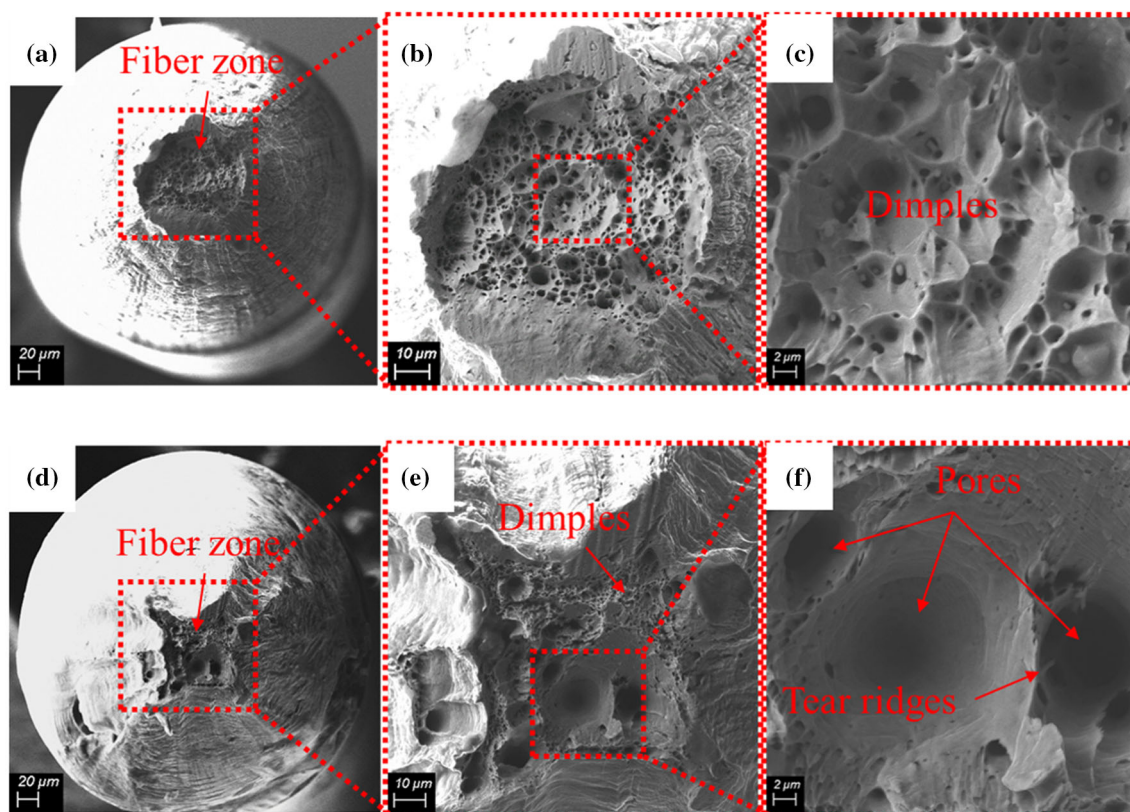


Fig. 10 Fracture surface morphology of two different offset conditions in NiTi-Cu weld joints (a), (b) and (c) centerline (0 μm) and (d), (e) and (f) 100 μm on Cu

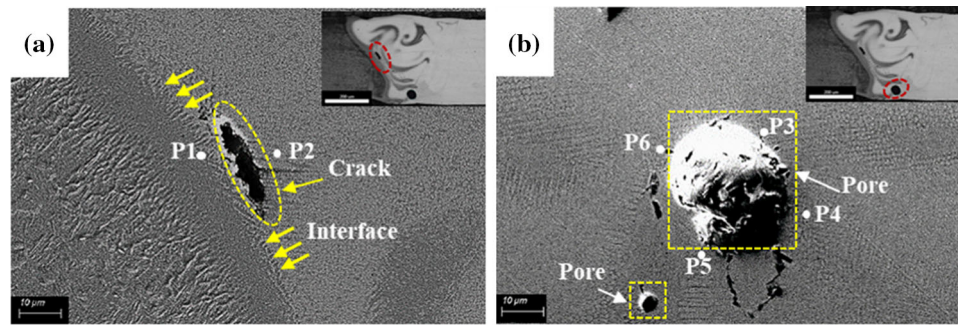


Fig. 11 Magnified microstructure of Cu (100 μm) joints showing (a) cracks and (b) pores inside the weld zone. Different points are indicated for EDS point scan analysis in Table 4

Table 4 Elemental distribution around cracks and pores in the Cu (100 μm) joint in selected points shown in Fig. 11

Points	Cu, at. %	Ni, at. %	Ti, at. %	Inferred phase (Ref 31-35)
P1	39.1	20.7	30.2	NiTiCu matrix with $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ IMCs Copper solid solution
P2	91.1	3.7	5.1	
P3	97.1	2.9	0	
P4	96.8	3.2	0	
P5	97.6	2.3	0	
P6	93.2	2.4	4.5	

4. Conclusions

NiTi shape-memory alloy and Cu wires were laser welded in butt joint configuration with different laser positioning to understand the effect of laser location on the microstructure and properties of these joints. The main conclusions were as follows:

- (1) Positioning the laser on the NiTi (50 μm) and the centerline (0 μm) resulted in a homogenous microstructure including the presence of $\text{Ni}_x\text{Ti}_y\text{Cu}_z$ intermetallic compounds. By changing the position onto the Cu (50 and 100 μm), the microstructure displayed complex flow and mixing patterns containing NiTiCu, pure NiTi, Cu solid solution and pure Cu. XRD measurements confirmed that the amount of NiTi and NiTiCu alloys decreased and pure Cu inside the weld zone increased, respectively.
- (2) Hardness profile maps that were in agreement showed an increase in hardness with increasing mixing of NiTi and Cu in the weld zone of the joints. By positioning the laser on the Cu, soft pure Cu and hard intermetallics compound inside the weld zone coexisted with a softened HAZ in the Cu side. However, the average hardness of the weld zone varied from ~ 512 to ~ 158 HV by positioning on NiTi and Cu, respectively.
- (3) Only the 100- μm Cu offset joint fractured in the weld zone during tensile loading, while the other joints failed in the Cu copper base material. The 100- μm Cu offset joint contained inhomogeneous microstructures, composition distributions, cracks and pores. Therefore, it was acceptable to weld NiTi and Cu wires with a 0-50- μm laser offset on the NiTi side.

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